

Ujjwal Basumatary

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Education

Indian Institute of Science, Bangalore
Bachelor and Master of Science, Major in Physics

Nov 2020 – Apr 2025
Physics: 9.6/10 (Overall: 9.3/10)

Publications

- [2410.22702](#) *Phys. Rev. D* 111, L041306
Beyond Hawking evaporation of black holes formed by dark matter in compact stars, Ujjwal Basumatary, Nirmal Raj and Anupam Ray.

Preprints

- [2512.02706](#)
Lectures on Quantum Field Theory on a Quantum Computer, Aninda Sinha and Ujjwal Basumatary.

Employment

Indian Institute of Science, Bangalore
Project Associate - 1

Jun 2025 – Jul 2025

Raman Research Institute, Bangalore
Project intern

Aug 2025 – Jun 2026
National Quantum Mission

Teaching assistantships

HE381: Quantum Field Theory and Quantum Computing
Instructor: Prof. Aninda Sinha

IISc, Bangalore
CHEP

- Conducted tutorials and designed/graded assignments, with demonstrations ranging from basic quantum algorithms to advanced Matrix Product States (MPS) algorithms. This course has led to a set of [lecture notes](#).
- The implementation details of the quantum and MPS algorithms are in [this](#) GitHub repository.

Fellowships and Awards

S. N. Bhatt Memorial Fellowship
International Center for Theoretical Sciences (ICTS)

May 2023 – Jul 2023
Bangalore

Kishore Vaiyanik Protsahan Yojana (KVPY) Fellowship
Department of Science and Technology (DST)

Nov 2020 – Apr 2025
India

Skills

- **Programming:** Proficient in C, C++, Python, shell script, Mathematica and Julia. Some experience with Haskell.
- **Markup:** Proficient in $\text{\LaTeX}$, working knowledge of HTML and CSS
- **OS:** Windows, GNU/Linux.

Advanced Graduate Coursework

Non-eq. Many-Body Dynamics	10/10	$\text{AdS}_3/\text{CFT}_2$	9/10
Standard Model	10/10	Adv. Mathematical Physics	10/10
QFT - 2	9/10	Cosmology for Theorists	9/10
Statistical Field Theory	9/10	QFT - 1	10/10
General Relativity	10/10	Nuclear and Particle Physics	10/10